

Bachelor's thesis

HMM for HFT

Predicting and trading on short-term regimes in currency tick data

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Abstract

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1 Introduction

1.1 The quantitative revolution

Modern financial markets are increasingly mathematical objects. What was once organized around human brokers, trading pits, voice execution, and discretionary judgement has gradually become an ecosystem of electronic limit-order books, automated market makers, statistical arbitrage engines, and latency-sensitive execution algorithms. This transformation is often described through the rise of high-frequency trading, but its deeper significance is the replacement of narrative-based market interpretation with probabilistic modelling, signal extraction, optimization, and real-time statistical decision-making. In this setting, prices are not merely economic summaries of fundamentals; they are also high-dimensional data streams generated by the interaction of heterogeneous agents, algorithms, inventories, order flows, and information shocks.

The popular account of this transition often begins with the drama of speed: faster cables, co-location, microwave links, and the race to react to information milliseconds before competitors (Lewis 2014; Aldridge 2013). However, from a mathematical perspective, speed is only one dimension of the problem. The more fundamental issue is that financial markets produce noisy, non-stationary time series in which any exploitable structure is weak, transient, and rapidly arbitrated away. This makes quantitative trading a natural domain for applied probability, statistics, stochastic processes, and machine learning. The trader is no longer simply asking whether an asset is “cheap” or “expensive”, but whether a small statistical regularity can be detected, validated, traded, and risk-managed after costs.

A useful symbolic figure in this quantitative revolution is Jim Simons, whose background in mathematics and codebreaking helped define the modern idea of finance as a signal-processing problem rather than a purely economic forecasting problem. The lesson is not that markets are deterministic or easily predictable, but almost the opposite: if markets are noisy, adaptive, and adversarial, then any predictive edge must be discovered through large-scale data analysis, rigorous out-of-sample testing, and careful modelling of uncertainty. This philosophy is closely related to the theme of this project. Instead of trying to forecast long-run currency fundamentals, we study whether short-term statistical structure in currency tick data can be captured by models that explicitly allow the market environment to change through time.

1.2 The microstructure of FX and market makers

The foreign exchange market is one of the most natural arenas in which to study this problem. FX is globally traded, highly liquid, electronically intermediated, and active nearly twenty-four hours a day. At the daily or weekly horizon, exchange rates are often modelled as approximately continuous stochastic processes driven by macroeconomic information, interest-rate expectations, risk sentiment, and international capital flows.

At the tick level, however, the market looks very different. Prices move in small discrete increments; bid and ask quotes update asynchronously; liquidity appears and disappears; and the observed price path becomes a sequence of small jumps, reversals, and microstructural frictions rather than a smooth curve.

This microscopic view is the domain of market microstructure. The central question is no longer only what the “true” value of a currency is, but how trading mechanisms, inventory constraints, asymmetric information, and order flow jointly determine observed prices (O’Hara 1995; Hasbrouck 2007). A market maker, for example, typically provides liquidity by quoting both bid and ask prices, earning compensation through the spread while managing inventory and adverse-selection risk. In a quiet regime, the market maker may be able to provide liquidity profitably as prices oscillate locally. In a stressed or information-driven regime, however, the same strategy may become dangerous: order flow can become one-sided, spreads may widen, and price changes may become directional rather than mean-reverting.

This distinction is crucial for high-frequency modelling. At lower frequencies, Brownian motion and diffusion models offer a powerful first approximation to asset-price dynamics, and the connection between finance and physics becomes visible through random walks, diffusion equations, and stochastic calculus (Bachelier 1900; Black and Scholes 1973; Merton 1973). At the tick level, however, the idealized diffusion picture becomes contaminated by the machinery of trading itself. Bid–ask bounce, discreteness, latency, varying liquidity, and order-book dynamics create patterns that are not well captured by a single stationary Gaussian model. The empirical object studied in this thesis is therefore not a clean theoretical Brownian path, but a noisy microstructural signal in which the statistical properties of returns may change rapidly over time.

Currency pairs are especially interesting in this regard because they often have economic and geographical links. Pairs such as AUD/USD and NZD/USD are exposed to related regional and commodity-market forces; EUR/NOK and EUR/SEK connect two Nordic currencies against the euro; and GBP/USD and EUR/USD share the US dollar leg and are deeply embedded in global macro trading. These relationships make currency pairs natural candidates for statistical arbitrage and pairs trading: rather than taking an outright directional view on a single exchange rate, one can model the relative movement between two related exchange rates and test whether temporary deviations tend to revert.



Figure 1: High frequency microstructure

1.3 Non-stationarity and market regimes

The main statistical difficulty is non-stationarity. A standard time-series model usually assumes that the same parameters govern the full sample, or at least that the estimated relationship is stable enough to remain useful out of sample. In financial markets this assumption is often too strong. Volatility clusters, correlations rise and fall, liquidity varies intraday, and the same trading rule may behave very differently depending on whether the market is calm, trending, news-driven, or illiquid. This is not merely an inconvenience; it is one of the defining mathematical features of financial data (Engle 1982; Cont 2001).

For pairs trading, non-stationarity is particularly damaging. A spread that appears mean-reverting in one period may drift in another. A hedge ratio estimated on a long historical sample may be irrelevant over a shorter local window. A pair may be statistically related in the full sample but too slow-moving to trade after transaction costs, or it may exhibit brief episodes of local mean reversion that are invisible to a static model. Thus, the empirical challenge is not only to detect dependence between two currency pairs, but to determine when this dependence is strong enough, stable enough, and cheap enough to trade.

This motivates the language of market regimes. Informally, a regime is a persistent market state with its own statistical properties. One regime may be characterized by low volatility, tight spreads, and local mean reversion; another by high volatility, weak pair-dependence, and directional price movement. The regime itself is not directly observed. We observe only the emitted data: returns, spreads, rolling volatility, correlations, and trading outcomes. The modelling problem is therefore naturally probabilistic: infer the hidden state from the observable time series, and use that inference to adapt the trading rule.

1.4 The solution in Hidden Markov Models

Hidden Markov Models provide a mathematically structured way to represent this idea. The model assumes that the observed process is driven by an unobserved Markov chain whose state determines the distribution of returns. The Markov assumption is deliberately simple: conditional on the current hidden state, the probability of the next state depends only on the present state and not on the full past. Despite this simplicity, the framework is powerful because it allows persistence, sudden switching, and regime-dependent parameters within a single probabilistic model (Rabiner 1989; Hamilton 1989).

In this project, the Hidden Markov Model is used as a regime filter for high-frequency currency-pair trading. The baseline model captures simple spread-based trading behaviour, while the autoregressive extension allows short-term serial dependence to enter the signal. The Markov-switching autoregressive model then allows the parameters of the return process to depend on the inferred market regime. Intuitively, the model asks whether there are hidden “safe” and “dangerous” states in the tick data, and whether avoiding the dangerous state improves the economic performance of a pairs-trading strategy.

This approach is attractive because it is interpretable. Unlike a black-box classifier, the HMM produces transition probabilities, regime-specific volatilities, regime-specific expected returns, and smoothed state probabilities. These quantities can be inspected directly and connected to financial intuition: a high-variance regime may correspond to unstable market conditions, while a low-variance regime may correspond to a more

favourable environment for mean reversion. In this sense, the HMM is not only a predictive model but also a diagnostic tool for studying how microstructural conditions evolve through time.

At the same time, the project is deliberately empirical and sceptical. The purpose is not to assume that a more complex model must outperform a simpler one. In high-frequency finance, additional model complexity can easily overfit noise, especially when trading costs, bid–ask spreads, and walk-forward validation are taken seriously. The key question is therefore comparative: does a Hidden Markov regime filter add value relative to simpler baseline and autoregressive trading rules, or does it mainly reduce exposure without improving risk-adjusted performance?

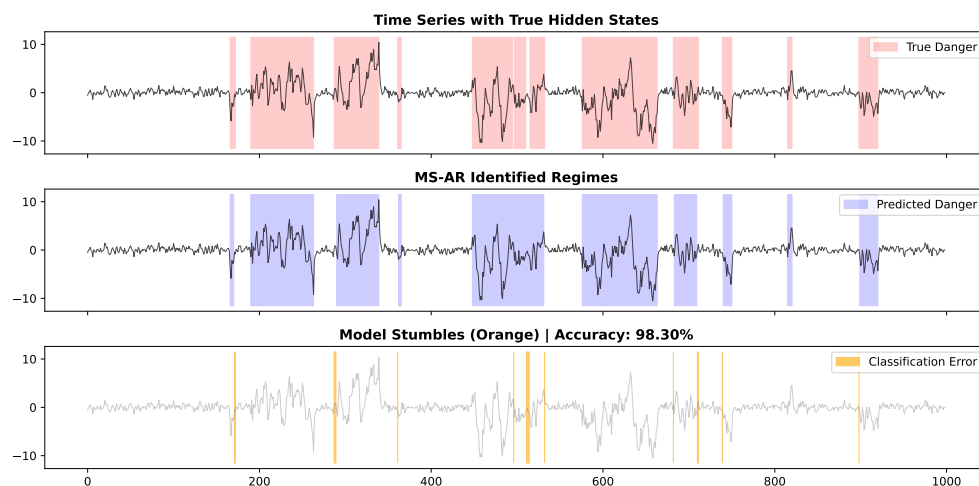


Figure 2: True and hidden states

1.5 Thesis structure

The remainder of the thesis is organized as follows. Section 2 introduces the statistical and financial methodology. It begins with the autoregressive benchmark model, then develops the Hidden Markov and Markov-switching autoregressive framework, before describing the data construction, model-selection procedure, transaction-cost assumptions, and backtesting metrics. Particular emphasis is placed on evaluating performance net of realistic trading frictions, since high-frequency strategies are highly sensitive to spreads and turnover.

Section 3 presents the empirical results for three currency-pair case studies: AUD/USD versus NZD/USD, EUR/NOK versus EUR/SEK, and GBP/USD versus EUR/USD. For each pair, the analysis studies liquidity, return distributions, pair-dependence, cointegration diagnostics, walk-forward optimization, and strategy performance. The baseline, autoregressive, and Markov-switching autoregressive strategies are compared in terms of total return, volatility, drawdown, Sharpe ratio, exposure, win rate, and tail behaviour.

Section 4 concludes by evaluating whether Hidden Markov Models are useful for high-frequency FX pairs trading. The central contribution of the thesis is not a claim that HMMs universally outperform simpler models, but a careful test of when regime awareness helps, when it fails, and how mathematical models behave when confronted with noisy, costly, non-stationary tick data.

2 Methods

In this section I introduce the methods. Namely, we start by introducing autoregressive models, before writing about Hidden Markov Models (HMM), introducing the data used in our simulations, while also presenting the statistical and financial metrics used to score the performance of our trading simulations.

2.1 Autoregressive models

When modeling time series data, a fundamental objective is to capture the linear dependence between current and past observations. The general Autoregressive model of order p , denoted as $AR(p)$, formalizes this intuition by projecting the current value of a variable Y_t as a linear combination of its p most recent past values:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \epsilon_t \quad (1)$$

In the context of financial returns, we establish our benchmark using the simplest form of this process: a single-regime autoregressive model of order one, or $AR(1)$. This model serves as a robust baseline because it efficiently captures short-term serial dependence, such as market momentum or immediate mean-reversion. The $AR(1)$ process for returns r_t is defined as:

$$r_t = c + \phi r_{t-1} + \epsilon_t \quad (2)$$

Here, c is the constant intercept (from which the unconditional mean return can be derived as $\frac{c}{1-\phi}$), ϕ measures the strength and direction of the linear serial dependence, and ϵ_t represents the stochastic innovation term, strictly assumed to be independent and identically distributed as Gaussian white noise, where $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$.

2.2 Hidden Markov Models

While the standard $AR(1)$ model provides a solid baseline, its primary limitation is the assumption of structural stability; it assumes that the parameters (c , ϕ , and σ^2) remain constant over time. Real-world financial markets, however, frequently undergo structural breaks and shift between distinct ‘regimes’—such as transitioning from a low-volatility bull market to a high-volatility bear market.

To capture this dynamic behaviour, we extend our framework using a Hidden Markov Model (HMM). We introduce an unobserved, discrete-time Markov chain, denoted by $S_t \in \{1, \dots, K\}$, which represents the active market regime at time t . The probability of the market transitioning from one state to another is governed by a set of transition probabilities:

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij} \quad (3)$$

where $p_{ij} \geq 0$ and the probabilities of transitioning from any state i to all possible states j sum to one.

$$\sum_{j=1}^K p_{ij} = 1 \quad \forall i \quad (4)$$

By conditioning our autoregressive process on this hidden state $S_t = k$, we allow the behavior of the returns to adapt to the current market environment. The regime-switching $AR(1)$ model is therefore defined as:

$$r_t = \mu_k + \phi_k r_{t-1} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_k^2) \quad (5)$$

This formulation is significantly more flexible than the single-regime benchmark. The parameter μ_k acts as the regime-specific drift, ϕ_k captures how the short-term memory of the market changes depending on the environment, and σ_k^2 models regime-specific volatility, effectively accommodating the volatility clustering frequently observed in financial time series.

To fully specify this Hidden Markov Model, we must estimate the complete parameter set θ , which encapsulates both the state-transition dynamics and the regime-dependent autoregressive parameters:

$$\theta = \{\pi, \mathbf{P}, \mu_k, \phi_k, \sigma_k^2 : k = 1, \dots, K\} \quad (6)$$

where $\pi = (\pi_1, \dots, \pi_K)^\top$ is the initial state distribution representing the probability $\mathbb{P}(S_1 = k)$ of starting in state k , $\mathbf{P} = (p_{ij})$ is the $K \times K$ transition matrix dictating the regime persistence, and the remaining terms represent the localized AR parameters for each respective state.

2.3 Model selection

After defining the structural framework of the Hidden Markov Model, the immediate empirical challenge is determining the optimal hyperparameters—most notably, the number of hidden states, K , and the autoregressive order, p . Adding more states or lags will invariably increase the model’s fit to the training data, but doing so introduces the severe risk of overfitting, where the model captures statistical noise rather than genuine market structure. To resolve this tradeoff, we evaluate competing models using Information Criteria, specifically the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC). Both metrics penalize models for unnecessary complexity while rewarding goodness-of-fit.

The AIC focuses on minimizing the information lost when a given model is used to approximate the true data-generating process, making it highly effective for optimizing predictive accuracy. It evaluates models based on m , which represents the total number of estimated parameters in the model, and $\hat{L}$, which is the maximized value of the likelihood function for the given data. The AIC is defined as:

$$\text{AIC} = 2m - 2 \ln(\hat{L}) \quad (7)$$

The BIC introduces a stricter penalty for model complexity by factoring in T , the total number of observations in the time series. Because the penalty term $\ln(T)$ scales with the sample size, the BIC aggressively penalizes overfitting in large financial datasets and mathematically favors the most parsimonious model that adequately explains the data. The BIC is defined as:

$$\text{BIC} = m \ln(T) - 2 \ln(\hat{L}) \quad (8)$$

During our hyperparameter sweep, models are estimated across a grid of potential K and p values. The configuration that minimizes the AIC and BIC is selected as the optimal structural architecture for our subsequent trading simulations.

2.4 Metrics

To rigorously evaluate the performance of the trading strategies generated in this simulation, we must rely on a robust set of quantitative metrics. Simply observing the final account balance is insufficient; a proper evaluation requires us to understand the path taken to achieve those returns, the risks assumed along the way, and the underlying mathematical behavior of the system. In this section, I introduce the comprehensive suite of metrics used to score the models. These indicators allow us to objectively compare different strategies, identifying their specific structural strengths, potential vulnerabilities, and overall viability in a live market context. Before detailing the specific calculations, we define the following standard notations used throughout this section:

- R_t : Return of the asset/strategy at time t
- μ : Mean return $\mathbb{E}(R)$
- σ Standard deviation of returns
- N : Total number of periods or trades
- V_t : Portfolio value at time t , and $P_t = \max_{s \leq t} V_s$ is the peak value up to time t

2.4.1 Statistical metrics

Statistical metrics form the foundational mathematical profile of our trading strategies. Rather than focusing directly on capital accumulation, these indicators analyze the shape, spread, and probability distribution of the underlying return data. By examining how returns behave on a probabilistic level, we can uncover hidden risks that standard equity curves might obscure—such as the likelihood of extreme market anomalies or structural asymmetries in the strategy’s outcomes. The following metrics are utilized to quantify the fundamental risk and distributional nature of the strategy:

Annual volatility quantifies the expected yearly fluctuation in returns, providing a baseline mathematical measure of standard risk.

$$\sigma_{\text{ann}} = \sigma \sqrt{T} \quad (9)$$

Skewness measures the asymmetry of the return distribution, helping to identify the likelihood of outsized positive or negative outliers.

$$S = \frac{E[(R - \mu)^3]}{\sigma^3} \quad (10)$$

Kurtosis measures the ‘fatness’ of the distribution tails, indicating the probability of rare, extreme market events.

$$K = \frac{E[(R - \mu)^4]}{\sigma^4} \quad (11)$$

Value at risk estimates the maximum expected loss over a specific time frame at a given confidence level α , acting as a standard gauge of the worst-case scenario.

$$\text{VaR}_\alpha = -Q_{1-\alpha}(R) \quad (12)$$

Conditional value at risk calculates the average expected loss assuming the VaR threshold is breached, providing a more comprehensive picture of extreme tail risk.

$$\text{CVaR}_\alpha = -E[R \mid R \leq -\text{VaR}_\alpha] \quad (13)$$

Tail factor quantifies the severity of tail risk by comparing the average extreme loss to the baseline extreme loss threshold.

$$\text{TF} = \frac{\text{CVaR}_\alpha}{\text{VaR}_\alpha} \quad (14)$$

2.4.2 Financial metrics

While statistical metrics describe the mathematical probability of returns, financial metrics translate those probabilities into practical, capital-based outcomes. These indicators evaluate the actual equity curve, measuring how efficiently a strategy deploys capital, how it navigates periods of historical loss, and its absolute trading profitability. Crucially, these metrics contextualize raw returns against the specific drawdowns and volatility endured to achieve them, providing a holistic view of the strategy’s execution quality and the psychological ‘holding stress’ it would impose on an investor. The core financial scoring metrics used in our evaluation are defined as follows:

Total return represents the overall cumulative growth or decline of the invested capital over the entire trading period.

$$\text{TR} = \prod_{t=1}^N (1 + R_t) - 1 \quad (15)$$

Annual return standardises the cumulative return into a yearly figure, allowing for direct comparison across different strategies and timeframes.

$$\text{AR} = (1 + \text{TR})^{\frac{1}{Y}} - 1 \quad (16)$$

Positive months counts the total number of profitable months, offering a straightforward measure of performance consistency over time.

$$N_{\text{pos}} = \sum_{t=1}^N \mathbb{I}(R_t > 0) \quad (17)$$

Max drawdown measures the largest peak-to-trough percentage drop in equity, revealing the strategy’s worst historical losing streak.

$$\text{MDD} = \max_t \left(\frac{P_t - V_t}{P_t} \right) \quad (18)$$

Max drawdown duration measures the longest time required to recover from a peak-to-trough loss, indicating capital recovery speed.

$$\text{MDD}_{\text{dur}} = \max(t_{\text{recovery}} - t_{\text{peak}}) \quad (19)$$

Ulcer index combines the depth and duration of drawdowns into a single composite metric to evaluate the holding stress of the investment.

$$\text{UI} = \sqrt{\frac{1}{N} \sum_{t=1}^N D_t^2} \quad (20)$$

Sharpe ratio measures the excess return generated per unit of total volatility, serving as the industry standard for risk-adjusted performance, where R_f is a ‘risk-free’ asset.

$$SR = \frac{\mu - R_f}{\sigma} \quad (21)$$

Sortino ratio modifies the Sharpe ratio to penalise only downside volatility, making it ideal for investors who view only negative fluctuations as true risk.

$$\text{Sortino} = \frac{\mu - R_f}{\sigma_d} \quad (22)$$

Calmar ratio evaluates the annualised return relative to the maximum drawdown, highlighting the specific tradeoff between reward and severe equity drops.

$$\text{Calmar} = \frac{AR}{MDD} \quad (23)$$

Profit factor shows the total money made for every dollar lost, acting as a pure, direct measure of absolute trading profitability.

$$PF = \frac{\sum \text{Gross Profits}}{\sum |\text{Gross Losses}|} \quad (24)$$

Payoff factor compares the absolute size of the average winning trade to the average losing trade, indicating the system’s risk-to-reward efficiency.

$$\text{Payoff} = \frac{\text{Average Win}}{|\text{Average Loss}|} \quad (25)$$

Win rate calculates the percentage of trades that result in a profit, reflecting the core accuracy of the strategy’s entry and exit signals.

$$W = \frac{N_{\text{wins}}}{N_{\text{trades}}} \quad (26)$$

Market exposure measures the fraction of time capital is actively deployed in the market as opposed to sitting safely in cash.

$$\text{Exposure} = \frac{T_{\text{invested}}}{T_{\text{total}}} \quad (27)$$

Average win calculates the typical gain realised on a successful transaction.

$$AW = \frac{1}{N_{\text{wins}}} \sum_{i=1}^{N_{\text{wins}}} \text{Profit}_i \quad (28)$$

Average loss calculates the typical capital deficit incurred on an unsuccessful transaction.

$$AL = \frac{1}{N_{\text{losses}}} \sum_{i=1}^{N_{\text{losses}}} \text{Loss}_i \quad (29)$$

2.5 Data

The empirical analysis is based on high-frequency foreign exchange data downloaded from Dukascopy. For each currency pair, I collect separate bid and ask quote data for both legs of the pair. The raw files are stored as monthly parquet files and loaded into Python through a reproducible data pipeline. The sample covers the period from January 2024 to December 2025, giving two full calendar years of tick-level FX observations.

The three currency-pair constructions used in the project are

$$\text{AUD/USD–NZD/USD}, \quad \text{EUR/NOK–EUR/SEK}, \quad \text{GBP/USD–EUR/USD}.$$

These pairs were selected because they represent economically related exchange rates while differing substantially in liquidity, transaction costs, and empirical dependence. AUD/USD and NZD/USD are both Asia-Pacific currencies quoted against the US dollar. EUR/NOK and EUR/SEK connect two Nordic currencies against the euro, but are less liquid and have materially wider spreads. GBP/USD and EUR/USD are highly liquid major currency pairs sharing the US dollar leg.

Using bid and ask data rather than only mid-prices is important in a high-frequency setting. At low frequencies, the spread may be treated as a small implementation detail, but at the tick level it is central to the economics of the strategy. A model may identify statistically significant mean reversion and still fail economically if the expected move is smaller than the cost of trading. For this reason, the data set is constructed to preserve bid prices, ask prices, mid-prices, and spread-based transaction-cost estimates throughout the pipeline.

2.6 Pre-processing

The raw Dukascopy quote data are not used directly in the statistical models. Instead, the data are transformed into synchronized pair-level observations through a custom Python pipeline. This preprocessing is important because high-frequency financial data are irregularly spaced, asynchronous, and strongly affected by bid–ask spreads and microstructure noise. The main preprocessing steps are described below.

2.6.1 Bid–Ask and Mid prices

For each currency leg, bid and ask quote streams are first sorted by timestamp. Since bid and ask updates do not necessarily arrive at exactly the same time, they are aligned using a backward as-of merge. This means that, at each timestamp, the most recent available quote is used, avoiding any use of future information.

The mid-price is then computed as

$$P_t = \frac{P_t^{\text{ask}}}{P_t^{\text{bid}}} \quad (30)$$

which produces a single representative price series for each asset while still preserving the original bid and ask quotes for transaction-cost calculations.

For asset $X \in \{A, B\}$ we synthesis a mid price at every tick

$$M_t^x = \frac{B_t^x + A_t^x}{2} \quad (31)$$

2.6.2 Spread

The bid-ask spread is central in HFT because expected profits are often small relative to trading costs. For each asset, the half-spread is measured in basis points as

$$\text{HS}_t = \frac{1}{2} \frac{P_t^{\text{ask}}}{P_t^{\text{bid}}} \times 10000 \quad (32)$$

The half-spread gives an estimate of the cost of crossing the market for one leg of the trade. Since pairs trading requires trading both legs, the combined spread cost is an important part of the later backtest.

Two non-stationary log-price series $\log A_t$ and $\log B_t$ are cointegrated if there exists a hedge ratio β such that the spread

$$S_t = \log A_t - \beta \log B_t \quad (33)$$

is stationary.

We model the cointegrating relationship as a slowly-varying linear regression of $\log A$ and $\log B$. We do not assume β is constant, we rather estimate β_t via rolling OLS with window size W_b .

$$\beta_t = \frac{W_b \sum_{s=t-W_b+1}^t x_s y_s - (\sum_s x_s)(\sum_s y_s)}{W_b \sum_s x_s^2 - (\sum_s x_s)^2} \quad (34)$$

$$\alpha_t = \frac{1}{W_b} \left(\sum_s y_s - \beta_t \sum_s x_s \right) \quad (35)$$

with $x_s = \log B_s$ and $y_s = \log A_s$

The cointegration spread at bar t is then

$$S_t = \log A_t - \beta_t \log B_t - \alpha_t \quad (36)$$

and the spread return is

$$\Delta S_t = r_t^A - \beta_{t-1} r_t^B \quad (37)$$

where $r_t^A = \log A_t - \log A_{t-1}$ etc

For the baseline strategy which knows nothing about regimes the trading signal is the rolling z-score of the spread

$$Z_t = \frac{S_t - \mu_t}{\sigma_t} \quad \mu_t = \frac{1}{W_z} \sum_{s=t-W_z+1}^t S_s \quad \sigma_t^2 = \frac{1}{W_z} \sum_{s=t-W_z+1}^t (S_s - \mu_t)^2 \quad (38)$$

with W_z bars.

2.7 Pairs trading

We choose to financial instruments, as AUDUSD and NZDUSD in this paper. Let us define the spread z_t as

$$z_t = p_t^{(1)} - \beta p_t^{(2)} \quad (39)$$

where β can be estimated by the Engle-Granger method, which means using an Ordinary Least Squares (OLS) regression followed by an Augmented Dickey-Fuller (ADF) test on the residuals to verify that z_t is stationary. It is also possible to estimate this directly in the HMM model, but easier to do so globally at first.

Thereafter we can fit an AR- or ARCH-HMM on the spread-process

$$z_t = \mu^k + \rho^k(z_{t-1} - \mu^k) + \varepsilon_t^k \quad (40)$$

where ρ^k measures how fast we expect the mean reversion to be in regime k . Values of ρ^k close to 0 indicate a faster mean reversion, while values closer to 1 indicate a slower mean reversion. Negative values of ρ^k would indicate even stronger mean reversion but possibly less realistic empirically.

The trading rule in pairs trading can be formulated as follows. In a drifting regime we do nothing. In a mean-reverting regime, say for regime $k = 2$ we trade by.

$$\begin{cases} z_t > \mu^2 + c\sigma^2 \Rightarrow \text{short AUDUSD and long NZDUSD,} \\ z_t < \mu^2 - c\sigma^2 \Rightarrow \text{long AUDUSD and short NZDUSD,} \\ z_t \approx \mu^2 \Rightarrow \text{close the position.} \end{cases} \quad (41)$$

While for one unit of AUDUSD we trade $\hat{\beta}$ units of NZDUSD, such that the position in main is exposed to the relative movement between AUDUSD and NZDUSD, and not toward the general level in the currency market.

For a soft HMM version the positions can be scaled with the probabilities of the model being in a certain mean-reverting regime

$$w(s_t) = \mathbb{P}(s_t \text{ is in a mean-reverting regime} \mid \mathcal{F}_{\sqcup}) \quad (42)$$

It may also be natural to scale the positions with estimated volatility.

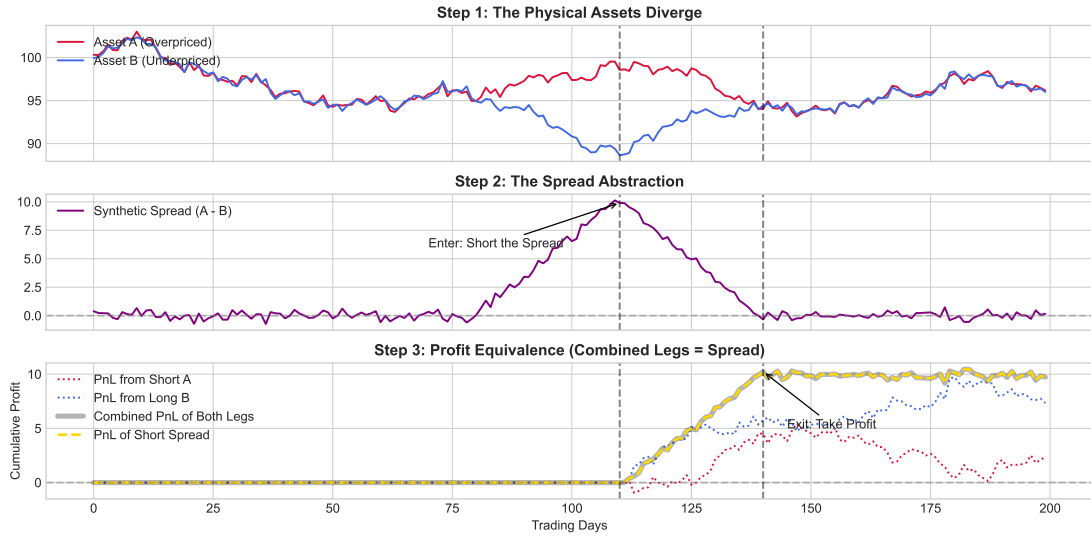


Figure 3: Long-short strategy in pairs trading

2.8 Back-testing

Back-testing is used to evaluate whether the model signals translate into economically meaningful trading performance. This is important because a model can fit historical returns well without producing a tradable strategy. In high-frequency trading, the expected profit per trade is often small, so even modest execution costs can eliminate an apparent statistical edge.

For this reason, the backtest includes realistic market frictions. The most important is the bid–ask spread: a trader generally buys at the ask and sells at the bid, rather than trading for free at the mid-price. Commissions and other execution costs are also included, since frequent position changes can make small costs accumulate quickly. Financing and margin costs are relevant when the strategy holds short or leveraged positions.

The strategies are evaluated in a walk-forward structure, where parameters are estimated on past data and then applied to later out-of-sample periods. This avoids look-ahead bias and gives a more realistic test of whether the baseline, AR, and Markov-switching strategies would have worked under implementable trading conditions.

2.9 Models

2.9.1 The Buy & Hold model

The do-nothing strategy, the position is always long the spread.

$$P_t^{\text{BH}} = +1 \quad \text{for every bar} \quad (43)$$

and it earns ΔS_t every bar and pays *no transaction costs*.

The purpose of the Buy & Hold is diagnostic, if $\sum_t \Delta S_t > 0$ for a given month, the spread drifted in our favour and a trading strategy needs to do better than that passive drift to add value, and if $\sum_t \Delta S_t < 0$ the spread drifted against us.

2.9.2 The Baseline model

The textbook z – score strategy, with two parameters, z_q = “quiet entry threshold” and z_x = “exit threshold”.

Position rules, applied bar-by-bar:

$$P_t = \begin{cases} +1 & \text{if } P_{t-1} = 0 \\ \text{and } Z_t < -z_q & \\ -1 & \text{if } P_{t-1} = 0 \\ \text{and } Z_t > +z_q & \\ 0 & \text{if } P_{t-1} = +1 \\ \text{and } Z_t \geq -z_q & \\ 0 & \text{if } P_{t-1} = -1 \\ \text{and } Z_t \leq -z_q & \\ P_{t-1} & \text{otherwise} \end{cases} \quad (44)$$

2.9.3 The Autoregressive (AR) model

Here we use the same entry/exit *direction* rules as the Baseline model with one extra binary condition, the gate

$$G_t = 1\{\pi_t^{\text{MR}} \geq 1 - \delta\} \quad (45)$$

where δ is the danger threshold parameter. The modified ruled is therefore

$$P_t = \begin{cases} 0 & \\ \text{if } G_t = 0 (\text{panic liquidation, no new entires}) & \\ \text{(same as baseline)} & \\ \text{if } G_t = 1 & \end{cases} \quad (46)$$

The kill switch is binary, we either trade normally or sit in cash, with no middle ground. Moreover the kill applies to existing positions too, not just to new entries, if we are long the spread and the HMM suddenly flags danger, we close out and go flat at the next bar.

2.9.4 The Markov-Switching Autoregressive (MS-AR) model

Let us denote S_t as the spread and $K_t \in \{1, 2\}$ a latent discrete regime indicator. The joint generative model is

$$S_t | S_{t-1}, K_t = k \sim N \left(c^{(k)} + \rho^{(k)} S_{t-1}, (\sigma^{(k)})^2 \right) \quad (47)$$

$$P(K_t = j | K_{t-1} = i) = p_{ij} \quad P = \begin{pmatrix} p_{11} & p_{12} & p_{21} & p_{22} \end{pmatrix} \quad p_{i1} + p_{i2} = 1 \quad (48)$$

This is an AR(1) on the spread with all three parameters switching, the constant $c^{(k)}$, the AR coefficient $\rho^{(k)}$ and the innovation variance $(\sigma^{(k)})^2$.

The unconditional mean of regime k when $|\rho^{(k)}| < 1$ is

$$\mu^{(k)} = \frac{c^{(k)}}{1 - \rho^{(k)}} \quad (49)$$

We fit the maximum likelihood with the EM algorithm, which alternates between the E-step, where we compute regime posteriors $\gamma_t^{(k)} = \mathbb{P}(K_t = k | S_1, \dots, S_t)$ via the forward-backward algorithm. And the M-step, where we update the regime-specific parameters $(c^{(k)}, \rho^{(k)}, \sigma^{(k)})$ and the transition matrix P to maximise the expected complete-data log-likelihood given the posteriors.

Winsorisation, we clip S_t at $\text{median}(S) \pm 4 \times \text{std}(S)$ before fitting. Also we scale by multiplying S_t by 10^4 before fitting.

The regime classification labelling is done as either

$$\text{MR index} = \arg \max_k \sigma^{(k)} \quad (50)$$

$$\text{DR index} = \arg \max_k \sigma^{(k)} \quad (51)$$

The in-sample regime probabilities are then found by

$$\pi_t^{\text{MR}} = \gamma_t^{(\text{MR})} = \mathbb{P}(K_t = \text{MR} | S_1, \dots, S_T) \quad (52)$$

$$\pi_t^{\text{DR}} = 1 - \pi_t^{\text{MR}} \quad (53)$$

Same entry/exit direction rules as Baseline, but the entry threshold itself is dynamic and probability-weighted

$$z_q^{\text{eff}}(t) = \pi_t^{\text{MR}} \cdot z_q + \pi_t^{\text{DR}} \cdot z_v \quad (54)$$

where z_v is the “volatile entry threshold”. The position rule is as follows

$$P_t = \begin{cases} +1 & \text{if } P_{t-1} = 0 \\ \text{and } Z_t < -z_q^{\text{eff}}(t) & \\ -1 & \text{if } P_{t-1} = 0 \\ \text{and } Z_t > +z_q^{\text{eff}}(t) & \\ 0 & \text{if } P_{t-1} \neq 0 \\ \text{and } |Z_t| \leq z_x & \\ +1 & \text{otherwise} \end{cases} \quad (55)$$

2.9.5 Model summary

Table 1: Strategy definitions and HMM usage.

Strategy	Always long?	Entry threshold (E_t)	Gate (G_t)	Reads HMM?
Buy & Hold	Yes	—	—	No
Baseline	No	z_q	1	No
AR	No	z_q	$\mathbb{1}\{\pi_t^{\text{MR}} \geq 1 - \delta\}$	Yes, binary
MS-AR	No	$\pi_t^{\text{MR}} z_q + \pi_t^{\text{DR}} z_v$	1	Yes, continuous

3 Results

4 Conclusion

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5 Appendix